Windows Event Logs and Wazuh Log Collection

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1 Introduction

Logs are one of the most valuable sources of information in cybersecurity. Every action performed by users, applications, services, or the operating system generates records that help security analysts understand what happened, when it happened, and how it happened. These records are essential for monitoring systems, detecting attacks, investigating incidents, and performing forensic analysis.

In Microsoft Windows environments, the Windows Event Log service provides centralized logging for system, application, and security events. When integrated with a Security Information and Event Management (SIEM) platform such as Wazuh, these logs can be collected, analyzed, and correlated to identify suspicious activities in real time.

This document introduces Windows Event Logs, explains their importance, demonstrates how to access them, and describes how Wazuh collects and analyzes these logs. The concepts presented in this report were learned through practical experimentation and a Windows Event Logging tutorial.

2 Logs and Events

Before analyzing Windows logs, it is important to understand the difference between logs and events.

- **Log:** A timestamped record automatically generated by an operating system, application, or service.
- **Event:** A single occurrence or action performed by the system or a user.
- **Log vs Event:** An event is the activity that occurs, while a log is the permanent record of that event.

Examples include:

- User logins
- File creation
- Application crashes
- Windows updates
- Service startup and shutdown
- PowerShell execution

3 Importance of Windows Event Logs

Windows Event Logs provide detailed visibility into activities occurring within the operating system. They are one of the primary sources of information used by Security Operations Centers (SOCs) to detect threats and investigate incidents.

Windows Event Logs help security analysts to:

- Monitor user authentication.
- Detect failed login attempts.
- Track account creation and deletion.
- Monitor PowerShell activity.
- Identify application errors.
- Detect suspicious processes.
- Investigate security incidents.

When forwarded to Wazuh, these events become searchable and can automatically trigger alerts based on predefined detection rules.

4 Categories of Windows Event Logs

Windows stores different types of events depending on their purpose.

4.1 Core Categories

System Logs

System logs contain events generated by the Windows operating system and core services.

Examples include:

- Driver failures
- Service startup
- Hardware issues
- System shutdown
- Operating system errors

Application Logs

Application logs record events generated by installed software.

Examples include:

- Browser events
- Database errors
- Software crashes
- Application updates

Security Logs

Security logs record authentication and security-related activities.

Examples include:

- Successful logons
- Failed logons
- User account creation
- Password changes
- Account lockouts
- Privilege escalation

These logs are heavily used by Wazuh during threat hunting.

4.2 Additional Categories

Setup Logs

Setup logs contain events generated during software installation or Windows updates.

Forwarded Events

Forwarded Events contain logs received from remote Windows systems using centralized log forwarding.

5 Accessing Windows Event Logs

Windows Event Logs are stored in:

```
C:\Windows\System32\winevt\Logs
```

The files use the **.evtx** format.

5.1 Viewing Logs Using Event Viewer

Windows provides a graphical interface called **Event Viewer**.

To open Event Viewer:

1. Press **Win + R**.
2. Type:
   eventvwr.msc
3. Press **Enter**.

Inside Event Viewer, the following categories are available:

- Application
- Security
- Setup
- System
- Forwarded Events

5.2 Viewing Logs Using PowerShell

Windows Event Logs can also be viewed using PowerShell.

Example:

```
Get-EventLog -LogName Application
```

To access Security logs, PowerShell must be executed with Administrator privileges.

6 Components of a Windows Event

Each Windows event contains multiple fields that assist during investigations.

Timestamp

The exact date and time when the event occurred.

Event ID

A numerical identifier that represents a specific event type.

Event Level

Indicates the severity of the event.

Possible levels include:

- Information
- Warning
- Error
- Critical
- Success Audit
- Failure Audit

Source

Identifies the application or Windows component that generated the event.

Description

Provides a detailed explanation of what occurred.

7 Common Windows Security Event IDs

The following Event IDs are commonly monitored during security investigations.

User Account Management

Event ID	Description
4720	User account created
4726	User account deleted
4732	User added to local group
4733	User removed from local group

Authentication

Event ID	Description
4624	Successful logon
4625	Failed logon

These events were observed during the SSH brute force simulation performed in the SOC lab.

PowerShell

Event ID	Description
4104	PowerShell Script Block Logging

This event was generated after enabling Script Block Logging and executing PowerShell commands during the PowerShell monitoring exercise.

Process Creation

Event ID	Description
4688	New process created

Security

Event ID	Description
1102	Security log cleared

8 Centralized Log Collection

Collecting logs from a single machine provides limited visibility. A SIEM platform centralizes logs from multiple systems, allowing security analysts to investigate events from a single interface.

Benefits include:

- Centralized log storage
- Faster searching
- Event correlation
- Threat detection
- Incident investigation
- Compliance monitoring

Common SIEM platforms include:

- Wazuh
- Microsoft Sentinel
- Splunk
- Elastic Security

9 Collecting Windows Event Logs in Wazuh

In this project, Windows Event Logs were collected using the Wazuh Agent installed on the Windows 11 endpoint.

The Wazuh Agent continuously monitored the configured Windows Event Channels and securely forwarded events to the Wazuh Manager running on the Ubuntu Server.

The primary logs collected included:

- Security
- System
- Application
- Microsoft-Windows-PowerShell/Operational
- Microsoft-Windows-Windows Defender/Operational
-

These logs were configured through the **ossec.conf** file located at:

```
C:\Program Files (x86)\ossec-agent\ossec.conf
```

After modifying the configuration, the Wazuh Agent service was restarted to apply the changes.

10 Viewing Events in Wazuh

Once collected by the Wazuh Manager, Windows Event Logs became available inside the Wazuh Dashboard.

Navigate to:

Threat Hunting
The dashboard displayed Windows events collected from the endpoint, including:

- Successful logons
- Failed logons
- User account creation
- PowerShell execution
- Windows Defender detections
- File Integrity Monitoring alerts

These events could be filtered using search queries, Event IDs, or Wazuh rule IDs to assist during investigations.

11 Conclusion

Windows Event Logs provide critical visibility into operating system activities and play an essential role in security monitoring and incident response. By forwarding these logs to Wazuh, security analysts gain a centralized platform for monitoring authentication events, PowerShell activity, Windows Defender alerts, and other security-related events.

Throughout this project, Windows Event Logs were successfully collected by the Wazuh Agent and analyzed within the Wazuh Dashboard. This centralized logging approach enabled efficient threat hunting and provided the foundation for detecting attacks such as SSH brute force attempts, PowerShell execution, malware detection, and other security events performed throughout the SOC lab.